

Ahmad Hossein Yazdani

ML Systems Software Engineer

US Citizen

Virginia Tech

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Summary

ML Systems Software Engineer experienced in leading the design and implementation of scalable memory- and storage-aware infrastructure for pre-training/post-training/inference workloads in production cloud systems and High Performance Computing (HPC). Expert in debugging large-scale complex distributed systems both performance-wise and functionality-wise. Experienced in building experimental systems and agentic AI workflows to mitigate distributed memory and I/O bottlenecks across GPU clusters (>100) and parallel storage.

Education

- 2020-August 2026 **PhD, Computer Science**, *Virginia Tech*, Blacksburg, VA, US.
Advised by Dr Ali Butt
Thesis title: Application-Aware I/O Prediction and Optimization for HPC and AI Workloads
- 2025 **Masters of Computer Science (Conferred en route to PhD)**, *Virginia Tech*, Blacksburg, VA, US.
Advised by Dr Ali Butt
- 2015–2020 **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.

Work Experience

- August, 2020 – August, 2026 **Research Assistant at Distributed System and Storage Lab, Virginia Tech.**
- Led the design and implementation of **SODA-LLM**, a storage-aware tensor placement system for distributed LLM training (PyTorch Tensor Parallelism, Fully Sharded Data Parallelism, DeepSpeed ZeRO), reduced **end-to-end latency by 20%** and **memory usage by 5%** on LLaMA-3-8B and Qwen-2-1.5B
 - Extended **OmniFed**, a federated learning system framework, with support for **gradient compression** during the Federated gradient aggregation stage on top of a Ray-based cluster, improving cross-site data-transfer efficiency and reducing communication overhead by **20%** in federated training at HPC and cloud scale.
 - Contributed to **SHADE** (A collaboration between Virginia Tech and UVA), an input data caching system for distributed CV training for cloud-scale infrastructure; increased cache hit ratios by up to **4.5×** over LRU and optimized data throughput.
- Spring 2026 **Research Intern, Oak Ridge National Laboratory.**
- Led the design of **IO-SENSE**, a multi-agent AI system predicting I/O pattern shifts driven by **application semantic changes** with **90%** accuracy
- Summer 2024 **Research Intern at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship.**
- Evaluated Drishti, an HPC I/O recommendation tool, identifying false-positive warnings and deriving insights to improve recommendation accuracy.
- Summer 2023 **Research Intern at Lawrence Berkeley National Laboratory (LBNL), internship.**
- Analyzed I/O variability for HPC workloads (E3SM, LAMMPS) on Lustre, identifying bottlenecks in shared access, metadata contention, and OST data distribution.
- Summer 2021 **Research Intern at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group.**

- leveraged unsupervised learning to quantify and predict I/O patterns of HPC jobs using historical workload features (90% accuracy)

Computer skills

Languages	Python, C, C++, pthread
ML	PyTorch, DeepSpeed, Hugging Face Transformers, CNN, CUDA, distributed training, LLM inference, vLLM, activation checkpointing/offloading, FlashAttention/Triton, Ray, unsupervised learning, MPI, NCCL
Cloud / Infrastructure	Parallel File Systems, Slurm, Chameleon Cloud gRPC, Lustre, GPFS, Slurm, MapReduce, Redis, MPI, containers
Software Engineering	Data Structures, Algorithms, Object-Oriented Design, Design Patterns, Git, CI/CD, Agile Development, Code Reviews, Technical Documentation

Selected Publications

- [ICDCN'25] **Yazdani, Ahmad Hossein** et al. User-based i/o profiling for leadership scale hpc workloads. International Conference on Distributed Computing and Networking (ICDCN '25), New York, NY, USA, Jan. 2025. Association for Computing Machinery. URL <https://doi.org/10.1145/3700838.3700865>.
- [FAST'23] Redwan Ibne Seraj Khan and **Yazdani, Ahmad Hossein** et al. Shade: Enable fundamental cacheability for distributed deep learning training. Santa Clara, CA, US, Feb. 2023. USENIX Association. URL <https://www.usenix.org/conference/fast23/presentation/khan>.